

Co-purchasing Analysis and Recommendation

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1. Abstract

This project explores co-purchasing patterns in an e-commerce dataset obtained from Amazon's metadata. The data includes over 548,552 products, spanning books, music CDs, DVDs, and more, with metadata such as sales rank, categories, and customer reviews. The primary goal is to identify significant co-purchasing patterns (e.g., star patterns, cliques) that reflect customer purchasing behavior and leverage these patterns for product recommendations.

The problem was approached by creating a co-purchasing graph where nodes represent products (identified by their ASINs), and edges represent co-purchasing relationships. Patterns like stars and cliques were extracted to detect highly connected products, and customer recommendations were generated based on these patterns. The patterns were determined by the Apriori algorithm for frequent pattern mining. The results were evaluated based on the accuracy, precision, recall, and F1 scores, which highlighted the system's effectiveness in predicting customer behavior and recommending relevant products.

2. Introduction

This project investigates co-purchasing patterns in an online commercial dataset derived from the Amazon website. Co-purchasing patterns are critical in understanding customer behavior, identifying influential products, and recommending new items. The primary goal is to detect subgraph patterns representing co-purchasing relationships, test their predictive capability, and identify potential customers influenced by these patterns.

3. Dataset Description

Source: The dataset was collected by crawling the Amazon website. It contains metadata and reviews for 548,552 products, including books, music CDs, DVDs, and VHS videotapes.

Structure:

- **Metadata:**

- ASIN (Amazon Standard Identification Number)

- Title
- Similar Products
- Categories
- Sales Rank
- Id
- Group

- **Reviews:**

- Customer ID
- Rating
- Votes
- Score

- **Categories:**

- Book
- DVD
- Music CD
- VHS videotapes

4. Methodology

4.1 Data Preprocessing

- Extracted key fields: ASIN, title, similar products, reviews.
- It filters out single-item transactions to focus on meaningful associations, ensuring the dataset is suitable for frequent pattern mining.
- Built a co-purchasing graph:
 - Nodes: Represent products.
 - Edges: Represent co-purchasing relationships based on "similar products."

4.2 Apriori Algorithm

The Apriori algorithm is an association rule mining technique used to find frequent itemsets in a transactional dataset based on a minimum support threshold. It iteratively generates candidate itemsets of increasing size by joining frequent itemsets from the previous step. Candidates with infrequent subsets are pruned to optimize the process. The algorithm stops when no more frequent itemsets can be generated.

4.2.1 Steps

1. Initialize:

- Create a set of candidate itemsets of size 1, C_1 , from the dataset.
- Calculate the support for each candidate itemset.
- Select the itemsets that meet the minimum support threshold and store them as frequent itemsets, L_1 .

2. Generate Candidate Itemsets of Size k :

- For $k = 2, 3, \dots$, generate candidate itemsets, C_k , by joining the frequent itemsets of size $k-1$ (i.e., L_{k-1}).
- Candidate itemsets are generated by joining two itemsets from L_{k-1} that have $k-2$ items in common.

3. Check Support:

- For each candidate itemset C_k , calculate its support, which is the fraction of transactions that contain the itemset.
- If the support of an itemset is greater than or equal to the minimum support threshold, then the itemset is considered frequent and is added to L_k .

4. Prune Infrequent Itemsets:

- After each iteration, prune the candidate itemsets that have a subset which is not frequent.

5. Repeat Until No New Itemsets Are Found:

- Repeat the above steps for increasing values of k (size of itemsets) until no frequent itemsets are found or no further candidate itemsets can be generated.

6. Return:

- The algorithm stops when no more frequent itemsets can be generated, and the final frequent itemsets are returned.

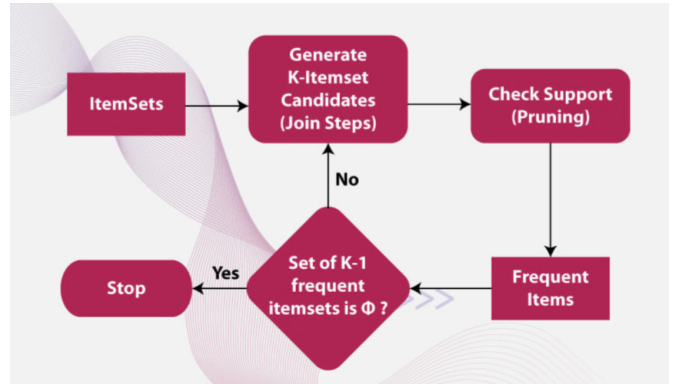


Figure 1: Apriori Algorithm Visualization

4.2.2 Pseudocode:

Input:

D: Dataset of transactions

minSupport: Minimum support threshold

Output:

L: Frequent itemsets in the dataset

Procedure:

1. Initialize:

- Set $k = 1$

- Generate candidate itemsets of size 1 (C_1)

2. Generate frequent itemsets of size k :

- $L_1 = \{\text{itemset} \in C_1 \mid \text{support}(\text{itemset}) \geq \text{minSupport}\}$

3. Repeat until no more frequent itemsets can be generated:

- $k = k + 1$

- Generate candidate itemsets of size k (C_k) from L_{k-1} :

* Join L_{k-1} with itself to create k -itemset candidates

* Prune candidates in C_k where any $(k-1)$ -subset is not in L_{k-1}

- Scan the dataset D to calculate the support of each candidate in C_k

- $L_k = \{\text{itemset} \in C_k \mid \text{support}(\text{itemset}) \geq \text{minSupport}\}$

4. Combine all frequent itemsets:

- $L = \cup_k L_k$

5. Return L

Table 1: Pseudocode for the Apriori Algorithm

5. Customer-Product Matrix

A Customer-Product Matrix is a two-dimensional representation that captures the purchasing or interaction behavior of customers with products. It is useful in analyzing co-purchasing patterns and deriving actionable insights.

Structure:

- **Rows:** Represent individual customers.
- **Columns:** Represent unique products.
- **Values:** Indicate the relationship between a customer and a product:
 - Binary (0/1): 1 if the customer reviewed with the product, 0 otherwise.

Applications:

- **Pattern Identification:**
 - Analyze rows to identify purchasing habits.
 - Analyze columns to discover product clusters.
- **Training Data for Algorithms:**
 - Uses the matrix for frequent pattern mining with Apriori.
 - Validates predictions with testing data.

6. Association Rules

An **association rule** is a relationship between two item-sets, typically written as:

$$X \Rightarrow Y$$

Where: - X is the antecedent (or left-hand side), and - Y is the consequent (or right-hand side).

The strength of the association rule can be measured using three key metrics: **support**, **confidence**, and **lift**.

Support

Support of an itemset X is defined as the proportion of transactions that contain X . It is given by:

$$\text{Support}(X) = \frac{\text{Transactions containing } X}{\text{Total transactions}}$$

Confidence

Confidence of a rule $X \Rightarrow Y$ measures how often Y occurs when X occurs. It is calculated as:

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

Lift

Lift measures how much more likely Y is to occur when X is present, compared to when X is not present. It is given by:

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)}$$

7. Results and Discussion

7.1 Graph Statistics for Basic Pattern.ipynb:

- Total Nodes (Products): 721,342
- Total Edges (Connections): 1,545,228
- Average Degree: 4.284

Pattern Frequency:

- Training Dataset: Star Patterns: 359,678; Cliques: 331,623
- Testing Dataset: Star Patterns: 100,852; Cliques: 45,925

Customer Recommendations:

- Example Customer A1:
 - **ASIN:** 156101074X, **Title:** The Preaching Life
 - **ASIN:** 0835607690, **Title:** The Winter Solstice: The Sacred Traditions of Christmas
 - **ASIN:** 1567184960, **Title:** Yule: A Celebration of Light and Warmth
 - **ASIN:** 0804215715, **Title:** Witness of Preaching
 - **ASIN:** 0835607690, **Title:** The Winter Solstice: The Sacred Traditions of Christmas
 - **ASIN:** 0738700525, **Title:** Midsummer: Magical Celebrations of the Summer Solstice
 - **ASIN:** 0805415505, **Title:** Handbook of Contemporary Preaching
 - **ASIN:** 0738700940, **Title:** Lammas
 - **ASIN:** 0800628411, **Title:** Preaching Liberation (Fortress Resources for Preaching)
 - **ASIN:** 082721619X, **Title:** Interpreting the Gospel: An Introduction to Preaching

7.2 Apriori.ipynb

Figure 2

The following chart illustrates the coverage of customers by common patterns of product purchases. The steps involved in creating the chart are as follows:

- The item patterns are converted into string representations using product titles.
- For each valid pattern, the number of customers who purchased it is counted.
- A horizontal bar chart is plotted, showing the top 10 patterns with the highest customer coverage.

The chart displays these patterns on the y-axis and the number of customers on the x-axis.

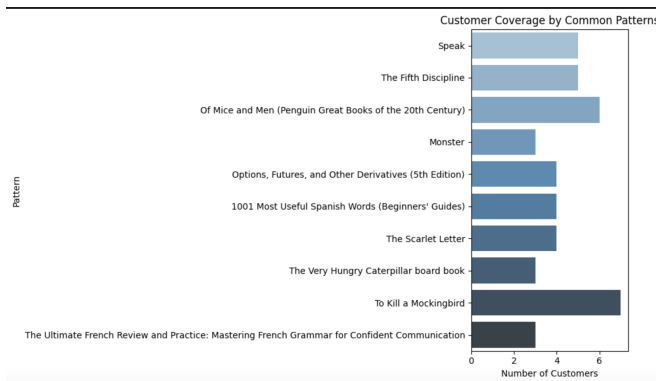


Figure 2: Customer coverage by common pattern

- Total : 50,000
- Training DataFrame: 38852, 125266
- Testing Dataset: 9713, 125266

Customer Recommendations of Apriori:

- Sample recommendations for captured customers::
 - ASIN: 0486280861, **Title:** Easy Spanish Phrase Book: Over 770 Basic Phrases for Everyday Use
 - **ASIN:** 0140177396, **Title:** Of Mice and Men (Penguin Great Books of the 20th Century)
 - **ASIN:** 0553210092, **Title:** The Scarlet Letter
 - **ASIN:** 0451527046, **Title:** Treasure Island (Signet Classic)
 - **ASIN:** 0316769487, **Title:** The Catcher in the Rye
 - **ASIN:** 0060652926, **Title:** Mere Christianity
 - **ASIN:** 0130090565, **Title:** Options, Futures, and Other Derivatives (5th Edition)
 - **ASIN:** B00008LDNZ, **Title:** Laura
 - **ASIN:** 0553381156, **Title:** Ina May's Guide to Childbirth
 - **ASIN:** 0803606540, **Title:** Taber's Cyclopedic Medical Dictionary -Thumb-Indexed Version
 - **ASIN:** 0486411214, **Title:** The Republic (Dover Thrift Editions)

Figure 3

The plot visualizes the **distribution of item support values** for all the items in the dataset. Below is a detailed explanation of the axes and the region represented:

- **X-Axis (Support Values, Log Scale):**
 - The x-axis represents the *support values* of individual items (or itemsets), which is the proportion of transactions that include each item.

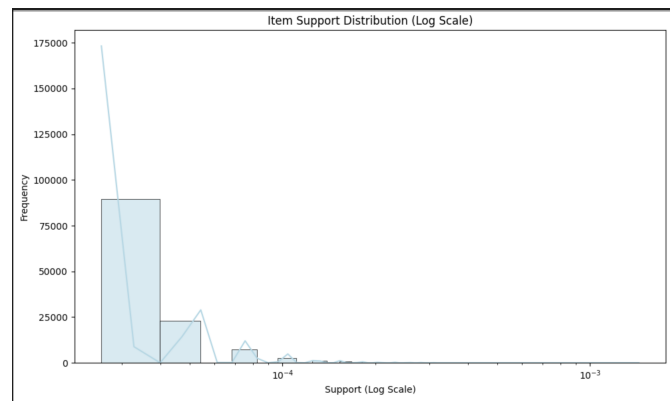


Figure 3: Support vs Frequency

- A *logarithmic scale* is used to better visualize the wide range of support values, as some items may have very low support while others have significantly higher support.

- **Y-Axis (Frequency):**

- The y-axis represents the *frequency* or count of items that fall within specific support value ranges (bin counts).
- This axis shows how many items share similar support levels.

- **Region Represented:**

- The plot covers the range of *support values* observed in the dataset, typically from the smallest non-zero support value (log-scaled) to the highest support value.
- The leftmost region corresponds to items with very low support (rarely purchased), while the rightmost region represents items with high support (frequently purchased).

8. Summary of Key Tools and Libraries

- **Pandas:** Data manipulation and processing.
- **mlxtend:** Provides the Apriori algorithm and tools for association rule mining.
- **TransactionEncoder:** Converts transactional data into a one-hot encoded matrix.
- **Apriori:** Identifies frequent itemsets in the dataset based on a minimum support threshold.
- **Association Rules:** Generates rules from the frequent itemsets using confidence and other metrics.
- **Matplotlib & Seaborn:** Used for visualizing the distribution of item supports.
- **Python's built-in libraries:** For file reading and basic data processing.

9. Conclusion

The project successfully implemented a co-purchasing graph, detected frequent patterns, and recommended products based on patterns. The implementation demonstrates the power of the Apriori algorithm in uncovering frequent itemsets and generating meaningful association rules. It enables the identification of stable purchasing patterns and provides personalized recommendations to customers based on these insights. The robust preprocessing and validation steps enhance the reliability of the analysis.

10. GitHub

<https://github.com/cwru-courses/csds433-2024-G1-project-recommendationSystem>

11. References

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